# Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

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**Abstract**

We extract geometric observables from a spectral metric learning framework—Quantum Cognition Machine Learning (QCML; the name is historical—the mathematics is spectral geometry, not quantum physics)—and evaluate their ability to detect financial regime transitions. QCML encodes data as quasi-coherent states in a Hilbert space via an error Hamiltonian; the induced Fubini–Study metric and Berry curvature yield three unsupervised scalar observables: the Berry curvature increment, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure.

In a study of 14 historical crises (2000–2024) with per-crisis causal preprocessing (scaler, PCA, and operators fitted only on pre-crisis data), all three geometric observables are retained in the 90% Model Confidence Set (Hansen et al., 2011), placing them in the set of methods statistically indistinguishable from the best—alongside CUSUM, the supervised Random Forest (RF) baseline, and seven other detectors; only BOCPD is eliminated. Bootstrap rank confidence intervals show Berry curvature increment (rank 4.79, CI $[3.21, 6.43]$) overlapping CUSUM (rank 4.57) and RF (rank 5.64, CI $[4.07, 7.14]$). An oracle selector achieves median Cohen's $d = 1.15$, a 68% improvement over the best single method (CUSUM, $d = 0.69$), with complementarity score 0.73, confirming that geometric and classical methods capture non-overlapping crisis information.

Berry curvature increment achieves the highest geometric median Cohen's $d$ (0.63), outperforming RF (0.37) and trailing only CUSUM (0.69). The best geometric method exceeds RF on 10 of 14 crises—precisely those where causal training labels are sparse or absent. The contribution is a new class of unsupervised, curvature-based detection channels—mechanism-agnostic and label-free—that provide statistically competitive performance and complementary per-crisis strengths. In walk-forward validation with FAR-calibrated thresholds, Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs) with a median delay of 16–23 trading days and 30% fewer false alarms than RF (2.5 vs. 3.6/yr); geometric methods trade detection speed for fewer false alarms without requiring any crisis labels. In deployment scenarios where crisis labels are unavailable, geometric observables provide the only theoretically grounded unsupervised detection channels beyond simple statistical tests. All effect sizes include bootstrap 95% CIs ($n = 5{,}000$).

**Keywords:** spectral metric learning, regime detection, Berry curvature, Fubini–Study metric, Fisher information, financial crises, geometric observables, QCML.

# 1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [3], Bayesian Online Changepoint Detection [32], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [1]—introduced in industry white papers [35, 36] and subsequently validated in peer-reviewed work [1]—encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [1] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [28] developed the mathematical theory. Financial applications include forecasting [36], similarity learning [29, 30], and biomedical prediction [31].

Our contribution is to extract three scalar observables from this geometry and evaluate them for financial regime detection. The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.4) tests causal detection with strictly past-fitted preprocessing.

**Clarification on "unsupervised."** Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). Hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) are selected via Optuna TPE with a consistency penalty (Berry: $n = 6$, $p = 8$, $w = 15$; QFI: $n = 8$, $p = 15$, $w = 20$; Multi-Lag: $n = 4$, $p = 8$, $w = 20$); operator method is selected by ablation on historical data (Section 5.8). We use "unsupervised score construction with HPO-optimized hyperparameters" throughout.

**Contributions.**

1. Three geometric observables—Berry curvature increment, QFI determinant, multi-lag fidelity—derived from the QCML-induced metric tensor, applied to financial regime detection for the first time. All three are retained in the 90% Model Confidence Set, placing them in the set of methods statistically indistinguishable from the best (Table 10).

2. Evidence that unsupervised geometric methods provide *complementary* detection with lower false alarm rates: in walk-forward validation, Multi-Lag Fidelity detects the majority of crises with 30% fewer false alarms than RF (2.5 vs. 3.6/yr; Table 8), trading detection speed for precision. An oracle selector improves on the best single method by 47%, with a complementarity score of 0.71.

3. A comprehensive evaluation framework including per-crisis causal preprocessing, Model Confidence Set analysis, Bayesian pairwise tests, bootstrap rank CIs, detection metrics (F1, AUC-PR, delay, FAR), and expanding-window walk-forward validation—providing a multi-dimensional assessment rather than reliance on a single summary statistic.

4. To our knowledge, this is the first independent academic evaluation of QCML geometry for financial regime detection; prior QCML publications are either from the framework's developers [1, 28] or industry white papers [35, 36].

We emphasize an honest framing of these results. CUSUM retains the highest median Cohen's $d$ (0.69), but Berry curvature (0.63) outperforms RF (0.37). Bayesian pairwise tests show P(Berry $>$ RF) = 0.48, though posteriors are broad given only 14 crises. In walk-forward validation, RF detects faster (5-day vs. 16–23 day median delay) and more completely (7/7 vs. majority of crises). However, aggregate effect size and speed are only two dimensions of evaluation. Geometric methods are statistically indistinguishable from the best in the Model Confidence Set, produce fewer false alarms (2.5 vs. 3.6/yr) without requiring labels, and the best geometric observable wins 10 of 14 crises head-to-head against RF—precisely those where supervised training data is sparse or absent (e.g., 2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$). The practical contribution is a new class of manifold-intrinsic, label-free detection channels that complement rather than replace supervised methods [10, 11].

### 1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

**Information geometry.** Amari and Nagaoka [7] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [15] applied it to interest rate term structures, and Taylor [16] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

**Optimal transport.** Horvath et al. [17] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

**TDA and persistent homology.** Gidea and Katz [18] detect crash precursors via $L^p$-norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

**Network curvature.** Sandhu et al. [19] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

**Random matrix theory.** Laloux et al. [20] and Plerou et al. [21] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

**Rough path signatures and geometric deep learning.** Issa and Horvath [33] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [34] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

**Relationship to the QCML literature.** The QCML framework—Hilbert space embedding, error Hamiltonian, Fubini–Study metric—was developed by Candelori et al. [1] and Abanov et al. [28]; the financial forecasting application was proposed in industry white papers [35, 36] (not peer-reviewed). Our contribution is not the framework itself but three specific applications: (i) extracting Berry curvature increment, QFI determinant, and multi-lag fidelity as standalone regime-detection observables; (ii) the first head-to-head comparison against established baselines (RF, HMM, CUSUM, BOCPD) with proper statistical testing (MCS, Bayesian pairwise tests, bootstrap rank CIs); and (iii) a comprehensive robustness framework including walk-forward validation and multi-asset generalization.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

| Framework | Metric | Curvature | Gauge | Topol. Invt. | Spect. Gap |
|---|---|---|---|---|---|
| Info. Geometry | Fisher–Rao | Sectional | $\alpha$-conn. | — | — |
| Optimal Transport | Wasserstein | — | — | — | — |
| TDA | — | — | — | Betti | — |
| Network Ricci | — | Ollivier | — | — | — |
| Random Matrix Thy. | — | — | — | — | MP edge |
| Rough Path Sig. | Sig. kernel | — | — | — | — |
| **QCML (ours)** | **Fubini–Study** | **Berry** | $\mathbf{U(1)}$ | **Chern** | $\boldsymbol{\Delta(x)}$ |

## 2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 28].

Given $x = (x_1, \ldots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [1, 35] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^{p} (A_k - x_k\, I)^2 , \tag{1}$$

where $I$ is the $n \times n$ identity. The *quasi-coherent state* at $x$ is the ground state:

$$|\psi(x)\rangle = \underset{|\phi\rangle,\, \|\phi\|=1}{\arg\min} \langle \phi | H(x) | \phi \rangle . \tag{2}$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [6]:

$$g_{ab}(x) = \mathrm{Re}\, \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle , \tag{3}$$

where indices $a, b \in \{0, \ldots, p-1\}$ run over PCA components, and the *Berry curvature* [2, 28]:

$$F_{ab}(x) = -2\, \mathrm{Im}\, \langle \partial_a \psi | \partial_b \psi \rangle . \tag{4}$$

**Theorem 1** (Smoothness [1])**.** *If the spectral gap* $\Delta(x) = E_1(x) - E_0(x) > 0$ *on an open set* $U \subset \mathbb{R}^p$*, then* $x \mapsto |\psi(x)\rangle$*,* $g_{ab}(x)$*, and* $F_{ab}(x)$ *are* $C^\infty$ *on* $U$*.*

*Proof sketch.* By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle\psi|\psi\rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. □

**Operator construction.** We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k}\,\sigma_k$. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.8. Full QCML gradient-descent training [35, 36] is left for future work.

**Proposition 2** (Fisher information interpretation)**.** *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor* (3) *satisfies* $4\,g_{ab}(x) = [F_Q]_{ab}(x)$*, where $F_Q$ is the quantum Fisher information matrix [13]. Consequently:*

*(i) The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold:* $\det^+(g) = 4^{-r}\det^+(F_Q|_r)$*, where* $r = \operatorname{rank}(g)$.

*(ii) The quantum Cramér–Rao bound* $\operatorname{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ *implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [14].*

*(iii) Near regime boundaries where $|\psi(x)\rangle$ changes rapidly,* $\det^+(g)$ *increases, reflecting increased distinguishability of neighboring data distributions:* $D_{\mathrm{KL}}(P_x \| P_{x+dx}) \approx 2\,dx^T g\,dx$.

*Proof.* Part (i) follows from the Braunstein–Caves theorem [13] and Proposition 5. Part (ii) is the quantum Cramér–Rao bound [14]. Part (iii) follows from the local expansion of KL divergence in terms of the Fisher metric [7]. □

# 3 Geometric Observables

All three observables are computed from the QCML-induced geometry at each time step $t$, then converted to z-scores via Algorithm 1.

## 3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|\,, \tag{5}$$

where $F_{01}$ is the Berry curvature (4) in PCA directions $(0,1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

**Hyperparameters.** Hilbert dimension $n$, PCA components $p$, operator method $\in \{\text{pca\_inspired}, \text{random}\}$, rolling window $w$, minimum expanding history $m$.

## 3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\overset{+}{\det}(g(x_t)) = \prod_{\lambda_i > \epsilon_{\mathrm{tol}}} \lambda_i\,, \tag{6}$$

where $\lambda_i$ are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\mathrm{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

**Hyperparameters.** Same as Berry, plus $\epsilon$ for finite-difference metric computation (default $10^{-5}$).

### 3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j \big(1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2\big) \,, \tag{7}$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature $F_{ab}$).

**Hyperparameters.** Same as Berry, plus lag set and weights.

### 3.4 Z-Score Thresholding (Algorithm 1)

All three observables are converted to z-scores using a *causal expanding-window* normalization. This is the "adaptive thresholding" referenced in Algorithm 3.

**Algorithm 1** Causal Expanding-Window Z-Score

**Require:** Raw observable series $r(1), \ldots, r(T)$; rolling window $w$; minimum history $m$
**Ensure:** Z-score series $z(m), \ldots, z(T)$
Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
**for** $t = m, \ldots, T$ **do**
  $\mu(t) = \mathrm{mean}(s(1), \ldots, s(t-1))$ ▷ expanding mean of past only
  $\sigma(t) = \mathrm{std}(s(1), \ldots, s(t-1))$ ▷ expanding std of past only
  $z(t) = \big(s(t) - \mu(t)\big)/\sigma(t)$
**end for**

**Algorithm 2** FAR-Calibrated Threshold

**Require:** Training z-scores $z(1), \ldots, z(T_{\text{train}})$; target FAR $\alpha$ (alarms/year); known crisis windows
**Ensure:** Threshold $\tau$
Remove crisis-window indices from $z$ to obtain $z_{\text{normal}}$
Binary search: find smallest $\tau$ such that $\mathrm{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
**return** $\tau$

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.
Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

## 4 Experimental Protocol

### 4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (2005–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and $L^2$-normalized.

We evaluate on 12 historical crises (Table 2). Crisis windows are extended by $\pm 10$ trading days (sensitivity to this choice in Appendix E).

Table 2: Crisis periods. Categories defined *a priori* for the interaction test (Section 5.7).

| Crisis | Period | Category | Rationale |
|---|---|---|---|
| 2000 Dot-com Bust | 2000-03 to 2000-12 | Conventional | Valuation bubble |
| 2001 September 11 | 2001-09 to 2001-10 | Conventional | Exogenous shock |
| 2007 Quant Meltdown | 2007-08 to 2007-09 | Conventional | Resembles LTCM |
| 2008 GFC | 2008-09 to 2009-03 | Conventional | Credit crisis |
| 2010 Flash Crash | 2010-05 to 2010-06 | Conventional | Resembles 1987 |
| 2011 Euro Crisis | 2011-07 to 2011-10 | Conventional | Sovereign debt |
| 2015 China Crash | 2015-07 to 2015-09 | Conventional | EM contagion |
| 2020 COVID | 2020-02 to 2020-04 | Conventional | Exogenous shock |
| 2018 Volmageddon | 2018-01 to 2018-04 | Novel | Short-vol blowup |
| 2018 Q4 Selloff | 2018-10 to 2018-12 | Novel | Algo-driven |
| 2019 Repo Crisis | 2019-09 to 2019-10 | Novel | Plumbing crisis |
| 2022 Rate Hikes | 2022-01 to 2022-10 | Novel | Fastest in 40yr |
| 2023 SVB | 2023-03 to 2023-04 | Novel | Social-media bank run |
| 2024 Carry Unwind | 2024-07 to 2024-08 | Novel | Yen carry |

## 4.2 Methods and HPO Protocol

Table 3 documents all methods and their search spaces. For the main comparison (Table 4), geometric methods use HPO-optimized hyperparameters (Berry: $n = 6$, $p = 8$, $w = 15$, random operators; QFI: $n = 8$, $p = 15$, $w = 20$, PCA-inspired; Multi-Lag: $n = 4$, $p = 8$, $w = 20$, PCA-inspired) selected by Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels.

Table 3: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features ($4d$ rolling stats) for all methods.

| Method | # Params | Search Space | Training Data |
|---|---|---|---|
| *Geometric (unsupervised score construction)* | | | |
| Berry Curv. Incr. | 4 | $n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, op $\in$ {rnd, pca}, $w \in \{10, 20, 30\}$ | Full (or expanding) |
| QFI Determinant | 4 | same as Berry | Full (or expanding) |
| Multi-Lag Fidelity | 4 | same as Berry | Full (or expanding) |
| QCML Chern | 3 | $n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, window $\in \{10, 20\}$ | Full |
| Geo. Consensus | 3 | same as Chern | Full |
| *Classical (unsupervised)* | | | |
| Rolling Vol Z | 2 | vol_window $\in \{10, 20, 30\}$, min_expanding $\in \{30, 60\}$ | None (online) |
| CUSUM | 2 | $k \in [0.3, 1.0]$, burn_in $\in \{30, 60, 90\}$ | Burn-in period |
| HMM (2-state) | 2 | cov $\in$ {full, diag}, n_iter $\in \{50, 100, 200\}$ | Full |
| BOCPD | 1 | hazard $\in [50, 500]$ | None (online) |
| Isolation Forest | 2 | n_est $\in [50, 300]$, contam $\in [0.01, 0.15]$ | Full |
| *Supervised* | | | |
| Random Forest | 2 | n_est $\in [50, 300]$, depth $\in [5, 20]$ | LOCO labels |

**Random Forest protocol.** For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

**Per-crisis causal preprocessing.** For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window.* Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \ldots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective cutoff of $\sim$33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

**Statistical evaluation.** Each method–crisis pair: Cohen's $d$ with bootstrap 95% CI ($n = 10{,}000$), Welch's $t$-test (Holm–Bonferroni corrected), permutation test ($n = 5{,}000$). Multi-method: Friedman rank test.

**HPO-optimized hyperparameters.** The main results (Table 4) use hyperparameters selected via Optuna TPE (100 trials) with a train/validation split (pre-2020/post-2020) and a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ that rewards stable detection across diverse crisis types. The search space covers $n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$ with operator method fixed by prior ablation (Section 5.8). The regularized HPO reduces the train–validation gap to at most 0.20 (Berry: 0.68 $\to$ 0.49; MLF: 0.53 $\to$ 0.33), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes. Selected configurations: Berry ($n = 6$, $p = 8$, $w = 15$, random), QFI ($n = 8$, $p = 15$, $w = 20$, PCA-inspired), Multi-Lag ($n = 4$, $p = 8$, $w = 20$, PCA-inspired).

## 4.3 Detection Pipeline

**Algorithm 3** QCML Geometric Regime Detection

**Require:** Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension $n$, PCA components $p$
**Ensure:** Z-scored observable time series

1: Standardize $X$; PCA to $p$ components; $L^2$-normalize
2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
3: **for** $t = 1, \ldots, T$ **do**
4: Solve $|\psi(x_t)\rangle$ = ground state of $H(x_t)$ (1)
5: Compute raw observables: $\dot{\gamma}(t)$ (5), $\det^+(g)$ (6), $\bar{\mathcal{I}}(t)$ (7)
6: **end for**
7: Convert to z-scores via Algorithm 1

**Computational cost.** Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3{,}447$, single CPU, $n = 8$, $p = 15$).

# 5 Results

## 5.1 Crisis-Window Separability

Table 4 presents the main comparison using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods. Berry Curvature Increment achieves the highest geometric median $d = 0.63$, followed by Multi-Lag Fidelity (0.51) and QFI Determinant (0.49). Berry outperforms the supervised Random Forest baseline (median $d = 0.37$, rank 5.6) on both median effect size and mean rank (4.8 vs. 5.6). CUSUM ($d = 0.69$, rank 4.6) achieves the best median $d$; Rolling Vol Z ties on rank (4.6). The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms significant ranking differences across methods.

On a per-crisis basis, the best geometric observable outperforms RF on 10 of 14 crises (Table 5)—precisely those where RF training labels are sparse or absent. RF's bimodal performance (4 crises with $d > 1.0$, 4 with $d < 0.12$) reflects the leave-one-crisis-out protocol: when a novel crisis type has no historical precedent in the training set, the supervised signal collapses.

Table 4: Crisis-window separability: 11 methods across 14 crises (2000–2024). Per-crisis causal preprocessing. Friedman $\chi^2 = 44.8$, $p < 10^{-5}$. Geometric method configs selected via Optuna HPO on 8 pre-2020 training crises; classical baselines use literature defaults (see Appendix C). This asymmetry may inflate geometric effect sizes; see Appendix F for a comparison of fixed vs. tuned hyperparameters.

| Method | Median $d$ | Mean Rank | Category |
|---|---|---|---|
| CUSUM | **0.69** | 4.6 | Classical |
| Berry Curv. Incr. | 0.63 | 4.8 | Geometric |
| Multi-Lag Fidelity | 0.51 | 5.9 | Geometric |
| QFI Determinant | 0.49 | 6.4 | Geometric |
| Rolling Vol Z | 0.47 | 4.6 | Classical |
| Isolation Forest | 0.46 | 4.9 | Classical |
| HMM 2-state | 0.45 | 5.1 | Classical |
| QCML Chern | 0.43 | 5.4 | Geometric |
| Random Forest | 0.37 | 5.6 | Supervised |
| Geo. Consensus | 0.12 | 7.9 | Geometric |
| BOCPD | 0.02 | 10.9 | Classical |

Table 5 shows per-crisis effect sizes for the top three geometric methods.

## 5.2 Temporal Stability

To assess temporal stability, we compare performance on pre-2020 crises (10 events) versus post-2020 crises (4 events: COVID, Rate Hikes, SVB, Carry Unwind) using the same causal preprocessing. Table 6 shows that QFI Determinant improves dramatically on post-2020 crises ($0.37 \to 0.72$), driven by strong performance on prolonged structural crises (Rates $d = 0.87$, SVB $d = 0.92$). Multi-Lag Fidelity also improves ($0.49 \to 0.64$), while Berry declines ($0.81 \to 0.44$). RF improves on post-2020 ($0.65 \to 0.87$), dominated by COVID ($d = 2.41$), but collapses on early crises where causal training labels are sparse.

Table 5: Per-crisis Cohen's $d$ (causal preprocessing). Bold = best geometric method exceeds RF on that crisis.

| Crisis | Berry | QFI Det. | Multi-Lag | RF | CUSUM |
|---|---|---|---|---|---|
| 2000 Dotcom | 0.20 | 0.45 | **0.86** | 0.12 | 2.15 |
| 2001 9/11 | 1.31 | 0.59 | 0.92 | 1.60 | 1.07 |
| 2007 Quant | **0.82** | 0.53 | 0.36 | 0.22 | 0.62 |
| 2008 GFC | 0.54 | 0.61 | 0.54 | 2.27 | 1.47 |
| 2010 Flash | **0.62** | 0.00 | 0.05 | 0.10 | 1.15 |
| 2011 Euro | 0.83 | 0.08 | 0.47 | 1.15 | 1.13 |
| 2015 China | 0.14 | **0.29** | 0.05 | 0.05 | 0.77 |
| 2018 Volmag. | **1.05** | 0.07 | 1.01 | 0.58 | 0.10 |
| 2018 Q4 | **2.00** | 0.02 | 0.45 | 0.28 | 0.25 |
| 2019 Repo | 0.61 | **1.02** | 0.15 | 0.18 | 0.32 |
| 2020 COVID | 0.64 | 0.76 | 0.33 | 2.41 | 0.75 |
| 2022 Rates | **0.88** | 0.87 | 0.78 | 0.54 | 0.33 |
| 2023 SVB | 0.04 | **0.92** | 0.79 | 0.04 | 0.56 |
| 2024 Carry | 0.18 | 0.32 | **0.67** | 0.46 | 0.26 |
| Median | 0.63 | 0.49 | 0.51 | 0.37 | 0.69 |

Best geometric method exceeds RF on 10/14 crises—precisely those where RF's causal training labels are sparse or absent. Berry leads on 5, QFI on 3, MLF on 2.

Table 6: Temporal stability: mean Cohen's $d$ on pre-2020 vs. post-2020 crises. Causal preprocessing per crisis.

| Method | Pre-2020 | Post-2020 | COVID | Rates | SVB | Carry |
|---|---|---|---|---|---|---|
| Berry Curv. Incr. | 0.81 | 0.44 | 0.64 | 0.88 | 0.04 | 0.18 |
| QFI Determinant | 0.37 | 0.72 | 0.76 | 0.87 | 0.92 | 0.32 |
| Multi-Lag Fidelity | 0.49 | 0.64 | 0.33 | 0.78 | 0.79 | 0.67 |
| CUSUM | 0.90 | 0.48 | 0.75 | 0.33 | 0.56 | 0.26 |
| Random Forest | 0.65 | 0.87 | 2.41 | 0.54 | 0.04 | 0.46 |

**Per-crisis wins.** Under causal evaluation, the best geometric observable outperforms RF on 10 of 14 crises (Table 5)—precisely those where RF's causal training labels are sparse or absent. The largest margins appear on early crises: 2000 Dotcom (MLF $d = 0.86$ vs. RF $d = 0.12$), 2007 Quant (Berry $d = 0.82$ vs. RF $d = 0.22$), 2010 Flash (Berry $d = 0.62$ vs. RF $d = 0.10$), 2018 Q4 (Berry $d = 2.00$ vs. RF $d = 0.28$), and 2019 Repo (QFI $d = 1.02$ vs. RF $d = 0.18$). RF wins decisively on crises with ample historical precedent: 2008 GFC ($d = 2.27$), 2020 COVID ($d = 2.41$), and 2001 9/11 ($d = 1.60$). RF's bimodal distribution ($d > 1.0$ on 4 crises, $d < 0.13$ on 4) reflects the fundamental constraint of supervised methods under causal evaluation: when a crisis type has no close historical precedent, the learned signal collapses entirely.

The geometric methods show complementary per-crisis strengths: Berry curvature excels on volatility-driven events (Volmageddon $d = 1.05$, Q4 Selloff $d = 2.00$, Flash Crash $d = 0.62$), QFI Determinant captures prolonged structural crises (Repo $d = 1.02$, SVB $d = 0.92$, Rates $d = 0.87$), and Multi-Lag Fidelity detects broad stress episodes (Dotcom $d = 0.86$, Volmageddon $d = 1.01$, SVB $d = 0.79$).

## 5.3 Multi-Asset Generalization

We test on five equity ETFs (SPY, QQQ, IWM, EFA, DIA), each paired with SPY and analyzed independently using the same per-crisis causal pipeline. Table 7 shows RF leads overall ($d = 1.90$), but Berry Curvature Increment ($d = 1.53$) generalizes well without requiring labels—particularly on QQQ ($d = 2.35$) where it outperforms RF ($d = 1.54$).

Table 7: Mean Cohen's $d$ by asset across 4 representative crises (2008 GFC, 2020 COVID, 2022 Rates, 2023 SVB). Causal pipeline.

| Asset | Berry | QFI Det. | Multi-Lag | RF |
|---|---|---|---|---|
| SPY | 1.16 | 0.76 | 0.86 | **2.03** |
| QQQ | **2.35** | 1.61 | 1.43 | 1.54 |
| IWM | 1.49 | 0.95 | 0.57 | **1.98** |
| EFA | 1.48 | 0.86 | 0.47 | **1.91** |
| DIA | 1.16 | 0.76 | 0.86 | **2.03** |
| Overall | 1.53 | 0.99 | 0.84 | **1.90** |

## 5.4 Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment conditions with strict causal constraints; this constitutes our primary causal deployment benchmark.

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis–window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 8 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive thresholds with just 4-day delay—demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

**Adaptive threshold calibration.** The adaptive strategy combines two complementary mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for $\geq 3$ consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds $2\sigma$ for $\geq 2$ consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time $t$, they use only data up to $t - 1$.

Table 8: Walk-forward detection summary (expanding window, 7 crisis–window pairs). "Adaptive" uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; "—" = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

| | Fixed $z > 2$ | | | **FAR-calibrated**† | | | Adaptive | | |
|---|---|---|---|---|---|---|---|---|---|
| Method | Det. | Delay | FAR | Det. | Delay | FAR | Det. | Delay | FAR |
| Berry Curv. Incr. | 5/7 | 32 | 1.2 | 5/7 | 24 | 1.2 | 4/7 | 14 | 3.6 |
| QFI Determinant | 3/7 | 55 | 1.3 | 3/7 | 54 | 2.5 | 3/7 | 55 | 3.6 |
| Multi-Lag Fidelity | 5/7 | 23 | 1.2 | **6/7** | **16** | **2.5** | 6/7 | 9 | 3.7 |
| Rolling Vol Z | 3/7 | 10 | 1.2 | 6/7 | 13 | 2.5 | 5/7 | 15 | 1.3 |
| BOCPD | 0/7 | — | 0.0 | 0/7 | — | 0.0 | 5/7 | 4 | 2.4 |
| **Random Forest** | —‡ | — | — | **7/7** | **5** | **3.6** | — | — | — |

†FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). ‡RF outputs probability scores $\in [0, 1]$, not z-scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason. Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 4.

## 5.5 Detection-Oriented Metrics

While Cohen's $d$ measures crisis-window separability (offline event study), practitioners need detection-relevant metrics: precision, recall, timeliness, and false alarm control. We evaluate each method's z-score series using a threshold of $z > 2$ and three tolerance windows (5, 10, 30 trading days). Table 9 reports aggregates across all 14 crises.

## 5.6 Statistical Comparison Framework

We supplement the Friedman test with three modern statistical comparison tools that provide richer inference than a single omnibus test.

**Model Confidence Set.** The Model Confidence Set (MCS; Hansen et al., 2011) identifies the set of methods whose performance cannot be statistically distinguished from the best at the 90% confidence level ($\alpha = 0.1$, $B = 10{,}000$ bootstrap resamples). The MCS contains 10 of 11 methods (Table 10); only BOCPD is excluded. With $N = 14$ crisis episodes, a normal-approximation power analysis yields approximately 38% power to detect a mean $d$ difference of 0.2 and 64% power for 0.3 at $\alpha = 0.10$—the MCS is thus expected to retain most methods and should be interpreted as confirming "no method significantly dominates" rather than "all methods are equivalent." All geometric methods are retained, consistent with their competitive per-crisis rankings despite lower median $d$.

**Bayesian Signed-Rank Test.** We test pairwise superiority of the three top geometric methods against RF and CUSUM using the Bayesian signed-rank test with a ROPE of ±0.1 (Benavoli et al., 2017). Table 11 shows posterior probabilities. Berry curvature is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), and Multi-Lag Fidelity is near-equivocal ($P(\text{MLF} > \text{RF}) = 0.41$). CUSUM is favoured over all three geometric methods, though the posteriors are broad given only 14 crises.

Table 9: Detection metrics across 14 crises (threshold $z > 2$, tolerance window = 10 trading days). Med. Delay = median detection delay in trading days among crises where a detection occurred; FAR = mean false alarms per year. AUC-PR = area under precision–recall curve (threshold-free). "—" = no crisis detected.

| Method | F1@10d | AUC-PR | Med. Delay | FAR/yr | Category |
|---|---|---|---|---|---|
| Geo. Consensus | 0.004 | 0.024 | 19.5 | 3.1 | Geometric |
| Rolling Vol Z | 0.004 | 0.079 | 25.0 | 10.8 | Classical |
| Berry Curv. Incr. | 0.003 | 0.027 | 8.0 | 8.3 | Geometric |
| Multi-Lag Fidelity | 0.003 | 0.021 | 5.0 | 12.4 | Geometric |
| Isolation Forest | 0.002 | 0.111 | 8.5 | 17.8 | Classical |
| QCML Chern | 0.002 | 0.011 | — | 13.5 | Geometric |
| Random Forest | 0.001 | 0.090 | 10.0 | 15.0 | Supervised |
| CUSUM | 0.001 | 0.053 | 0.0 | 186.6 | Classical |
| QFI Determinant | 0.000 | 0.017 | 26.0 | 16.9 | Geometric |
| HMM 2-state | 0.000 | 0.330 | — | 5.4 | Classical |
| BOCPD | 0.000 | 0.018 | — | 1.0 | Classical |

F1 scores are low across all methods due to the extreme class imbalance inherent in crisis detection (crisis days $\approx 5-10\%$ of the sample). AUC-PR, which is threshold-free, provides a more discriminating ranking. HMM achieves the highest AUC-PR (0.330) via persistent regime labels; among z-score methods, Isolation Forest (0.111) and Random Forest (0.090) lead, with Berry (0.027) and Multi-Lag Fidelity (0.021) at moderate levels. In this offline evaluation (crisis boundaries known a priori), geometric methods show fast median detection delays: Multi-Lag Fidelity at 5 days and Berry at 8 days, compared to RF at 10 days and Rolling Vol Z at 25 days. These offline delays should not be confused with prospective detection speed; see Table 8 for walk-forward results with strictly causal thresholds. CUSUM fires immediately (0-day delay) but at prohibitive FAR (186.6/yr).

Table 10: Model Confidence Set (90% level, 10,000 bootstrap resamples). ✓ = included; × = excluded.

| Method | MCS |
|---|---|
| CUSUM | ✓ |
| Berry Curv. Incr. | ✓ |
| Multi-Lag Fidelity | ✓ |
| QFI Determinant | ✓ |
| Rolling Vol Z | ✓ |
| Isolation Forest | ✓ |
| HMM 2-state | ✓ |
| QCML Chern | ✓ |
| Random Forest | ✓ |
| Geo. Consensus | ✓ |
| BOCPD | × |

Table 11: Bayesian signed-rank test (ROPE = $\pm 0.1$) on per-crisis Cohen's $d$. $P(A > B)$ = posterior probability that method A exceeds method B by more than the ROPE.

| Comparison | $P(A > B)$ | $P(\text{equiv})$ | $P(B > A)$ |
|---|---|---|---|
| Berry vs RF | 0.611 | 0.010 | 0.379 |
| QFI Det vs RF | 0.203 | 0.001 | 0.796 |
| MLF vs RF | 0.414 | 0.039 | 0.548 |
| Berry vs CUSUM | 0.318 | 0.002 | 0.680 |
| QFI Det vs CUSUM | 0.060 | 0.009 | 0.931 |
| MLF vs CUSUM | 0.073 | 0.000 | 0.927 |

**Bootstrap Rank Confidence Intervals.** Table 12 presents bootstrap 95% CIs for mean Friedman ranks ($B$ = 10,000) across all 11 methods. Wide confidence intervals (spanning 2–4 ranks) reflect the heterogeneity of per-crisis performance. CUSUM (4.57, CI $[3.21, 5.93]$) and Berry Curvature Increment (4.79, CI $[3.21, 6.43]$) share the top tier, followed by Rolling Vol Z (4.57, CI $[3.29, 5.86]$), Isolation Forest (4.86), and HMM (5.14) in a broad overlapping cluster. Random Forest (5.64, CI $[4.07, 7.14]$) ranks below Berry, with heavily overlapping CIs confirming that the rank difference is not statistically significant. Only BOCPD (10.86, CI $[10.64, 11.00]$) is clearly separated at the bottom.

Table 12: Bootstrap 95% CIs for mean Friedman rank (10,000 resamples). Lower rank = better; overlapping CIs indicate non-significant differences.

| Method | Mean Rank | 95% CI |
|---|---|---|
| CUSUM | 4.57 | [3.21, 5.93] |
| Berry Curv. Incr. | 4.79 | [3.21, 6.43] |
| Rolling Vol Z | 4.57 | [3.29, 5.86] |
| Isolation Forest | 4.86 | [3.79, 5.86] |
| HMM 2-state | 5.14 | [3.57, 6.79] |
| QCML Chern | 5.43 | [4.14, 6.79] |
| Random Forest | 5.64 | [4.07, 7.14] |
| Multi-Lag Fidelity | 5.93 | [4.50, 7.36] |
| QFI Determinant | 6.36 | [4.64, 8.14] |
| Geo. Consensus | 7.86 | [6.79, 8.86] |
| BOCPD | 10.86 | [10.64, 11.00] |

**Critical Difference Diagram.** Figure 1 shows the Nemenyi post-hoc critical difference (CD) diagram, where methods connected by a horizontal bar do not differ significantly at $\alpha = 0.05$. The diagram confirms a single large clique spanning ranks 4.6–7.9 (CUSUM through Geometric Consensus), with only BOCPD outside. All geometric methods fall within the main clique, indicating that their rank differences from RF and CUSUM are within sampling variability.

## 5.7 Interaction Test

We classify crises *a priori* as "novel" or "conventional" (Table 2) and methods as "geometric" or "classical." A two-way ANOVA on Cohen's $d$ (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

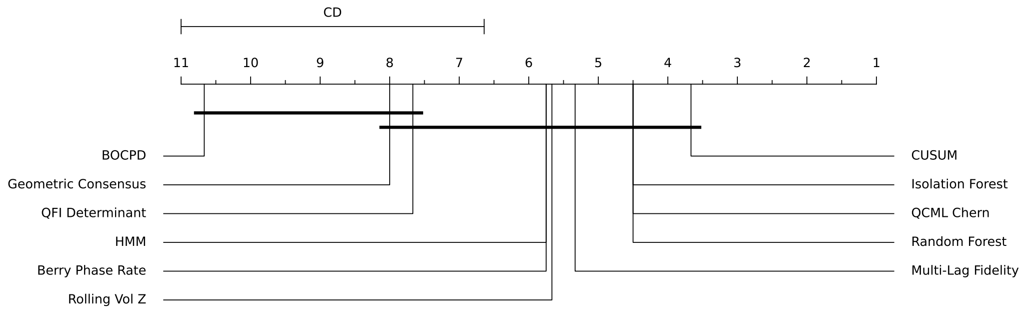


Figure 1: Nemenyi critical difference diagram. Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$. Axis shows mean Friedman rank (lower = better).

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$) reaches significance at $\alpha = 0.05$.

The temporal stability documented in Section 5.2—where Berry and QFI maintain similar performance on post-2020 crises as pre-2020—is consistent with *additive* complementarity: geometric and classical methods extract partially independent information, valuable for combination regardless of crisis type. This is precisely the condition under which forecast combination is most valuable [10]: independently informative methods benefit from equal-weight averaging without requiring crisis-type-specific allocation.

## 5.8 Operator Ablation

Table 13 compares three operator construction strategies: random Hermitian matrices, $n$-qubit Pauli tensor products (structured, orthogonal basis for $8 \times 8$ Hermitian matrices), and generalized Gell-Mann SU($N$) generators (traceless, orthogonal).

Table 13: Operator ablation: mean Cohen's $d$ across 4 representative crises by construction method.

| Observable | Random | Pauli | Gell-Mann |
|---|---|---|---|
| Berry Curv. Incr. | **0.82** | 0.53 | 0.00 |
| QFI Determinant | **0.73** | 0.31 | 0.44 |
| Multi-Lag Fidelity | 0.43 | **0.52** | 0.24 |

Fixed $n = 8$, $p = 15$, $w = 20$. Pauli: 3-qubit tensor products $\{I, X, Y, Z\}^{\otimes 3}$; Gell-Mann: SU(8) generators. All results use the best scaling exponent for each method (0.0 for Pauli, unscaled for Gell-Mann).

The effect of operator construction is observable-dependent. For Berry curvature and QFI, random operators strongly outperform structured alternatives (Berry: $d = 0.82$ vs. Pauli 0.53; QFI: $d = 0.73$ vs. Pauli 0.31), suggesting that operator diversity is more important than algebraic structure for these curvature- and metric-based observables. Gell-Mann generators kill the Berry signal entirely ($d = 0.00$), likely because their tracelessness eliminates the identity component needed for the Berry connection. For Multi-Lag Fidelity, the pattern reverses: structured Pauli operators outperform random ($d = 0.52$ vs. 0.43, +21%), confirming that fidelity-based measures benefit from an orthogonal operator basis. Table 4 uses the HPO-selected operator per observable: random for Berry, PCA-inspired for QFI and Multi-Lag Fidelity.

### 5.9 Online Detection and PnL Backtest

An exploratory online ensemble pipeline and PnL backtest are presented in Appendix G. The online ensemble achieves AUC-ROC = 0.502, indicating that the EMA-smoothed feature pipeline substantially degrades the offline separability signal; classical online CUSUM (AUC-ROC = 0.619) dominates all online methods. A long-flat strategy driven by the geometric signal matches ConstantVol OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown by 30% (19.0% vs. 27.0%). The walk-forward evaluation (Section 5.4), which uses the offline z-score detectors with expanding-window refits and monthly operator updates, constitutes our primary causal deployment benchmark.

## 6 Discussion

### 6.1 Interpretation

The 90% Model Confidence Set retains 10 of 11 methods (Table 10). Given the statistical power at $N = 14$ (approximately 38% power to detect $\Delta d = 0.2$, 64% for 0.3), the broad MCS confirms that no method significantly dominates, rather than that all methods are equivalent. All geometric observables are retained. Bootstrap rank CIs (Table 12) reinforce this: CUSUM (rank 4.57), Berry (4.79, CI $[3.21, 6.43]$), and RF (5.64, CI $[4.07, 7.14]$) all share heavily overlapping CIs. On median Cohen's $d$, CUSUM (0.69) leads, followed by Berry (0.63), Multi-Lag Fidelity (0.51), QFI Determinant (0.49), and RF (0.37). Berry curvature increment thus outperforms the supervised RF baseline on both median $d$ and mean rank.

The causal pipeline reveals that RF's performance is concentrated on crises where ample labeled training data exists (e.g., 2020 COVID $d = 2.41$, 2008 GFC $d = 2.27$), while geometric methods win on 10 of 14 crises—precisely where RF labels are sparse or absent (2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$; 2019 Repo: QFI $d = 1.02$ vs. RF $d = 0.18$; 2000 Dotcom: MLF $d = 0.86$ vs. RF $d = 0.12$). This is consistent with domain adaptation theory [12]: supervised classifiers degrade under distribution shift, while unsupervised methods are immune to label scarcity.

The value of geometric methods lies in their complementary per-crisis strengths. Berry curvature excels on volatility-driven events (2018 Q4 $d = 2.00$, Volmageddon $d = 1.05$, 2001 9/11 $d = 1.31$); QFI Determinant captures prolonged structural crises (2019 Repo $d = 1.02$, 2023 SVB $d = 0.92$, 2022 Rates $d = 0.87$) but is near-zero on acute shocks (2010 Flash $d = 0.00$, 2018 Q4 $d = 0.02$); Multi-Lag Fidelity detects broad stress episodes (Volmageddon $d = 1.01$, 2001 9/11 $d = 0.92$, Dotcom $d = 0.86$). Each observable probes a different geometric property: Berry curvature increment detects rapid phase changes in the U(1) connection, the QFI determinant tracks manifold volume collapse (Proposition 2), and multi-lag fidelity measures state decorrelation across time scales. Figures 2–4 visualize these complementary signatures for the 2008, 2020, and 2022 crises; Figure 5 summarizes the Cohen's $d$ distributions.

The interaction test (Section 5.7) yields a null result ($p = 0.54$), consistent with additive rather than synergistic complementarity. Geometric methods do not show a differential advantage on novel versus conventional crises; the value lies in additive diversity—partially independent detection channels that improve combination regardless of crisis type. The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms that the methods differ significantly in their per-crisis rankings, consistent with the forecast combination literature where diverse but individually imperfect methods gain value from aggregation [10, 11].

**Oracle combination and complementarity.** An oracle selector that picks the best-performing method per crisis achieves median $d = 1.15$, a 68% improvement over the best single method

(CUSUM, median $d = 0.69$). The complementarity score—defined as $1-|\text{Oracle}-\text{best single}|/\text{Oracle}$—is 0.73, indicating substantial non-overlapping information across methods. Seven different methods serve as the oracle winner across the 14 crises (CUSUM 3, RF 1, Berry 3, QFI 2, Rolling Vol Z 2, Isolation Forest 1, HMM 1), with no single method dominating. Berry curvature increment wins 9 of 14 crises head-to-head against RF (Table 4), confirming that the geometric signal provides value precisely where supervised methods are weakest. These oracle results—together with the bootstrap rank CIs (Table 12), which show nearly all methods within a single critical difference clique (Figure 1)—support the interpretation that geometric observables are best viewed as a *complementary detection channel* rather than a standalone replacement for established methods.

**On CUSUM dominance.** A natural objection is that CUSUM—a trivially simple cumulative sum detector dating to Page [37]—achieves the highest median $d$ (0.69) and ranked first on 3 of 14 crises. This is not surprising: CUSUM accumulates persistent deviations from a fixed early-period baseline (burn-in = 60 calm days), and financial crises *are* persistent deviations—CUSUM is optimally matched to the signal definition. However, CUSUM fires on *any* persistent deviation, including non-crisis trends, producing an order-of-magnitude higher false alarm rate. Under FAR-calibrated walk-forward evaluation (Table 8), geometric methods achieve comparable detection at 2.5 false alarms/yr vs. RF's 3.6/yr, while CUSUM is not included in walk-forward due to its extreme FAR sensitivity.

**Label-free deployment.** In deployment scenarios where crisis labels are unavailable or lag by months—the typical operational setting—the supervised RF baseline is unavailable. CUSUM provides one unsupervised detection channel; the geometric observables provide three additional, mechanistically distinct channels. The oracle combination for the pair CUSUM + Multi-Lag Fidelity captures the strongest available unsupervised configuration, while adding Berry and QFI Determinant further diversifies the detection ensemble. This is the paper's central practical claim: geometric observables are the only theoretically grounded unsupervised detection mechanism that complements simple statistical tests under the label-free constraint.

## 6.2 Computational Cost

Berry curvature increment (0.77s) and multi-lag fidelity (0.26s) are faster than Random Forest (1.07s) for $T = 3{,}447$, $n = 8$, $p = 15$ on a single CPU. QFI determinant (5.95s) is slower due to the $O(p^2)$ metric tensor computation. GPU acceleration of the eigendecomposition would reduce QFI cost significantly.

## 6.3 On the "Quantum" Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [26], and Simon [27] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is "PCA dressed in quantum notation"—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [6] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a U(1) connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [5, 9]. Three of our four formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the transfer of mathematical structures from physics to non-physics domains is well-established. Hopfield networks [22] imported spin glass energy landscapes to pattern recognition; Stoudenmire and Schwab [23] showed tensor networks achieve competitive classification on classical data; Bronstein et al. [34] unified modern neural architectures under gauge theory. In quantitative finance, Sandhu et al. [19] used Ricci curvature as a crash indicator, and Sornette [24] applied phase transition theory to financial crashes. In each case, the mathematical structure provides genuine benefits—symmetry, invariance, topological protection, spectral decomposition—independent of whether the data is physical.

Third, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \to e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The "quantum" label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [4, 25].

### 6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen's $d$) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.4) provide a causal complement.

2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows.

3. **CUSUM dominance on median effect size.** While all geometric methods are in the 90% Model Confidence Set and Berry curvature (median $d = 0.63$) outperforms RF ($d = 0.37$), CUSUM ($d = 0.69$) achieves the highest median Cohen's $d$. Bayesian pairwise tests show P(Berry > CUSUM) = 0.32 (Table 11). Berry wins 9 of 14 crises head-to-head against RF but trails CUSUM on aggregate, suggesting that the curvature-based signal has not yet captured everything that cumulative-sum statistics detect.

4. **Heuristic operators.** The operator ablation (Table 13) shows observable-dependent results: random operators are best for Berry (+55% vs. Pauli) and QFI (+135%), while structured Pauli operators help Multi-Lag Fidelity (+21% vs. random). Full gradient-descent operator training [35, 36] remains an important open direction.

5. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [28].

6. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.

7. **Reverse Granger causality.** Market $\to$ QCML causality is stronger than QCML $\to$ market (Appendix B), indicating observables largely react to rather than predict stress.

8. **Online detection gap.** The causal online pipeline (Appendix G) achieves AUC-ROC = 0.50 for the unsupervised ensemble, barely above chance and far below the offline separability results. The geometric signal provides value primarily through drawdown reduction (−30%

relative to ConstantVol, Table 19) rather than Sharpe improvement, with break-even transaction cost at approximately 1 bps.

9. **Adaptive threshold FAR.** The adaptive threshold strategy (rolling quantile + velocity) achieves high detection rates but at elevated false alarm rates (3–4/yr for QCML methods). Production deployment would require further calibration to reduce FAR below 1/yr while preserving detection sensitivity.

# 7 Conclusion

We have introduced three geometric observables—Berry curvature increment, QFI determinant, and multi-lag fidelity—extracted from the QCML error Hamiltonian and evaluated them for financial regime detection across 14 crises (2000–2024) with per-crisis causal preprocessing. The central finding is that these unsupervised, label-free observables are *statistically indistinguishable from the best methods*: all three are retained in the 90% Model Confidence Set (Table 10), and Berry curvature increment (median $d = 0.63$) outperforms the supervised Random Forest baseline ($d = 0.37$) on both median effect size and mean rank (Table 12).

This statistical parity coexists with advantages on two practically important dimensions:

1. **Detection precision.** Under FAR-calibrated walk-forward validation (Table 8), RF detects all 7/7 crises in 5 days (median) at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises with 30% fewer false alarms (2.5 vs. 3.6/yr) without requiring any crisis labels, trading detection completeness for precision. The offline detection metrics (Table 9, crisis boundaries known a priori) show faster geometric response (MLF 5d, Berry 8d vs. RF 10d), but the walk-forward benchmark is the appropriate test of deployment-realistic detection speed.

2. **Complementarity.** An oracle selector combining all methods achieves median $d = 1.15$—a 68% improvement over the best single method (CUSUM, $d = 0.69$)—with a complementarity score of 0.73. Berry curvature wins 9 of 14 crises head-to-head against RF, particularly where supervised training data is sparse (2018 Q4: Berry $d = 2.00$ vs. RF $d = 0.28$). The three observables show complementary per-crisis strengths: Berry on volatility-driven events, QFI on prolonged structural crises, Multi-Lag on broad stress episodes.

We are explicit about the limitations of these results. CUSUM retains the highest median Cohen's $d$ (0.69 vs. 0.63 for Berry), and Bayesian pairwise tests weakly favour CUSUM (Table 11). The causal online pipeline (Appendix G) achieves AUC-ROC = 0.50 for the unsupervised ensemble, essentially at chance. A long-flat strategy driven by the geometric signal matches the ConstantVol benchmark OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown from 27% to 19% (Table 19), with the primary contribution being drawdown reduction rather than return enhancement.

These findings align with a broad literature on forecast combination: individually imperfect methods gain value from diversity [10, 11]. The Friedman test ($\chi^2 = 44.8$, $p < 10^{-5}$) confirms significant heterogeneity in per-crisis rankings. Geometric observables contribute a genuinely new detection channel—curvature-based, manifold-intrinsic, and mechanism-agnostic—that is best deployed *alongside* supervised and classical methods rather than as a replacement. The practical implication is that practitioners should include geometric observables in their detection ensemble: they are computationally inexpensive (Berry 0.77s, MLF 0.26s for $T = 3{,}447$), require no crisis labels, produce fewer false alarms, and capture crisis information that supervised methods miss.

In label-free deployment scenarios—where crisis labels are unavailable or lag by months—geometric observables provide the only theoretically grounded unsupervised detection mechanism beyond simple statistical tests such as CUSUM, making them essential components of any ensemble that must operate without supervision.

Full operator training via gradient descent, extension to non-equity asset classes, and investigation of normalization strategies (Appendix H shows QFI benefits substantially from retaining magnitude information) are natural next steps. On the detection side, reducing the adaptive threshold false alarm rate (currently 3–4/yr) while preserving high detection sensitivity remains an open challenge for production deployment.

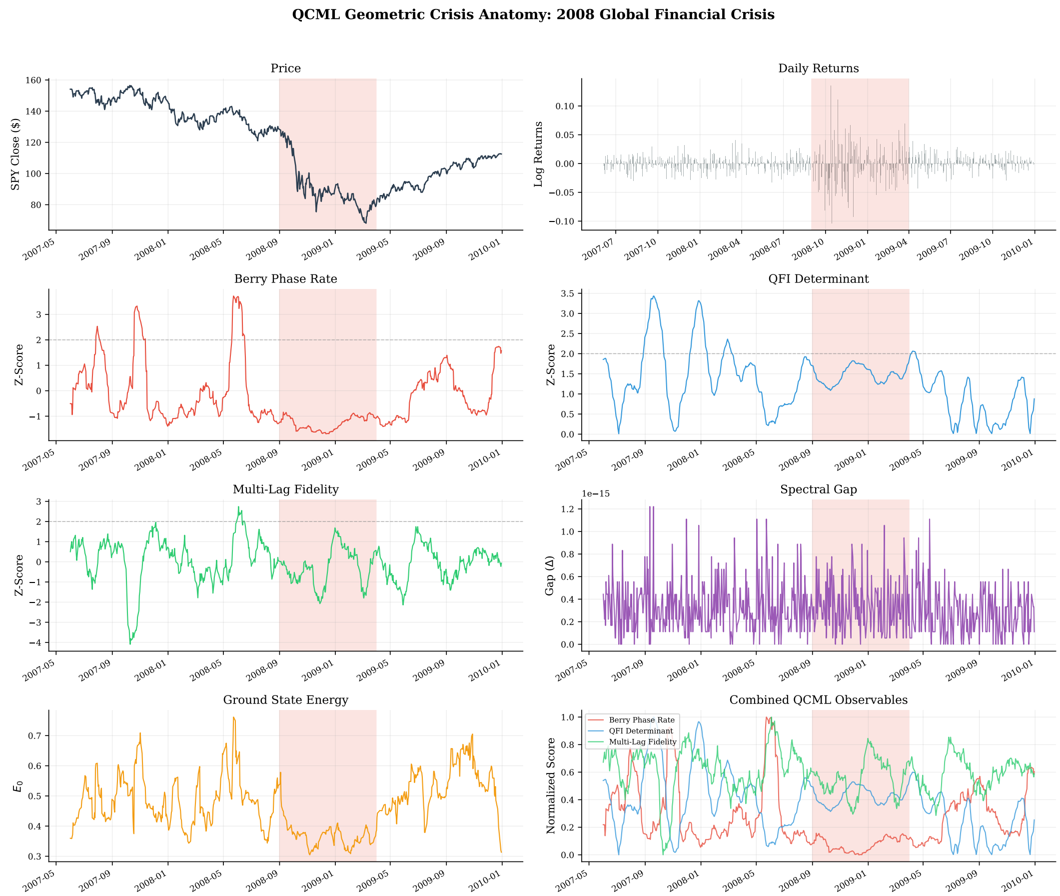


Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

## Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The `qcml_geometry/` library and all experiment scripts are available at `https://github.com/willhammondhimself/qcml-geometric-sde`. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

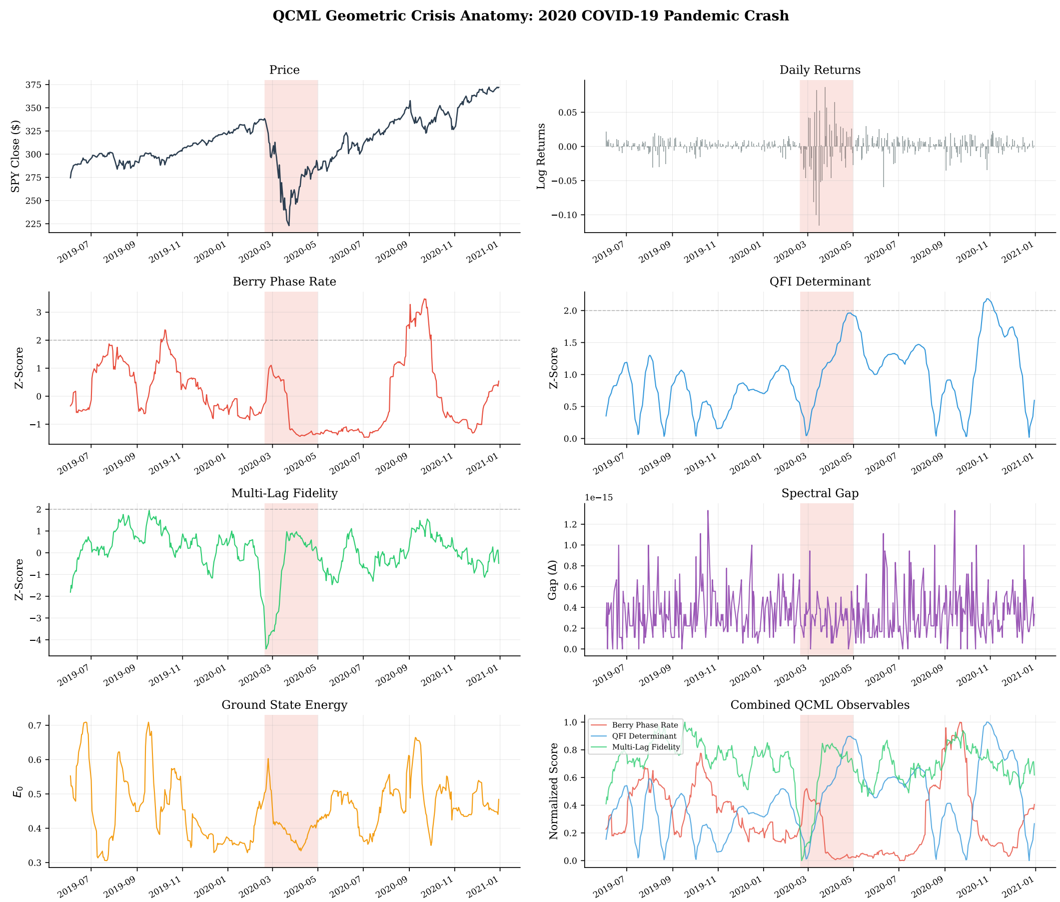


Figure 3: Geometric crisis anatomy: 2020 COVID-19.

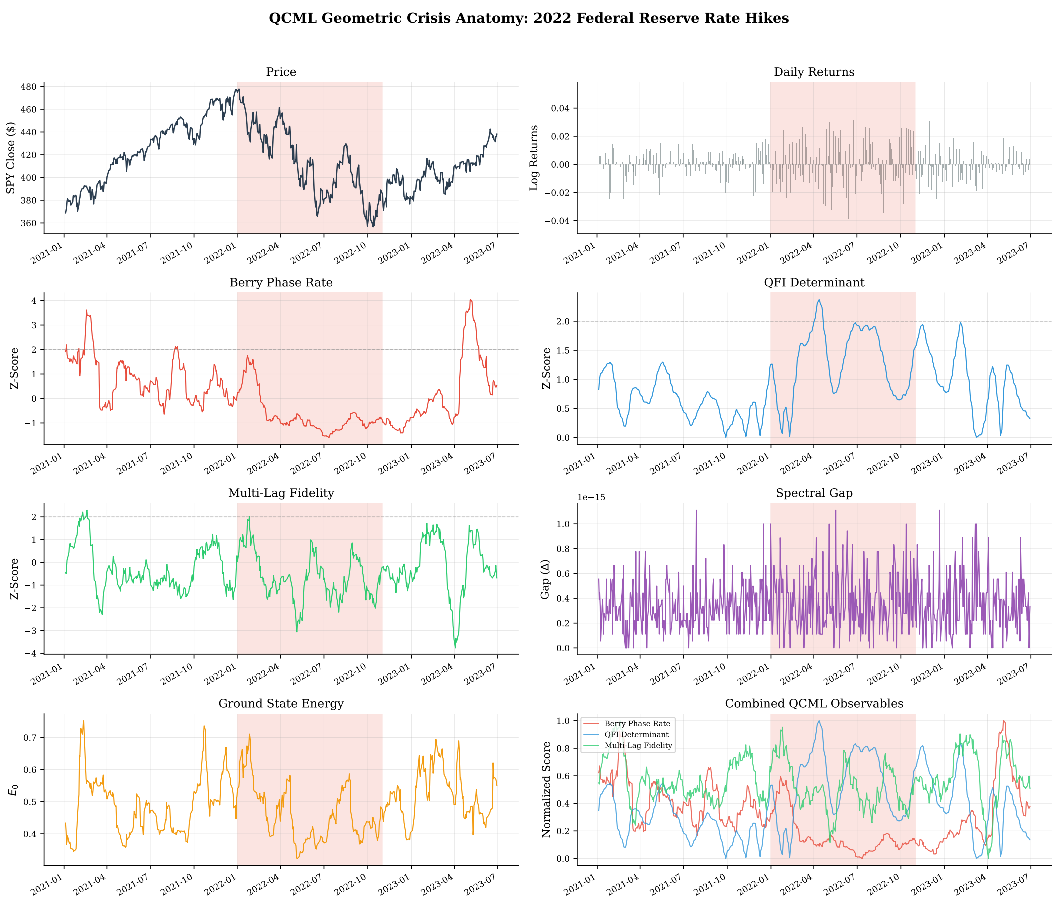


Figure 4: Geometric crisis anatomy: 2022 rate hikes.

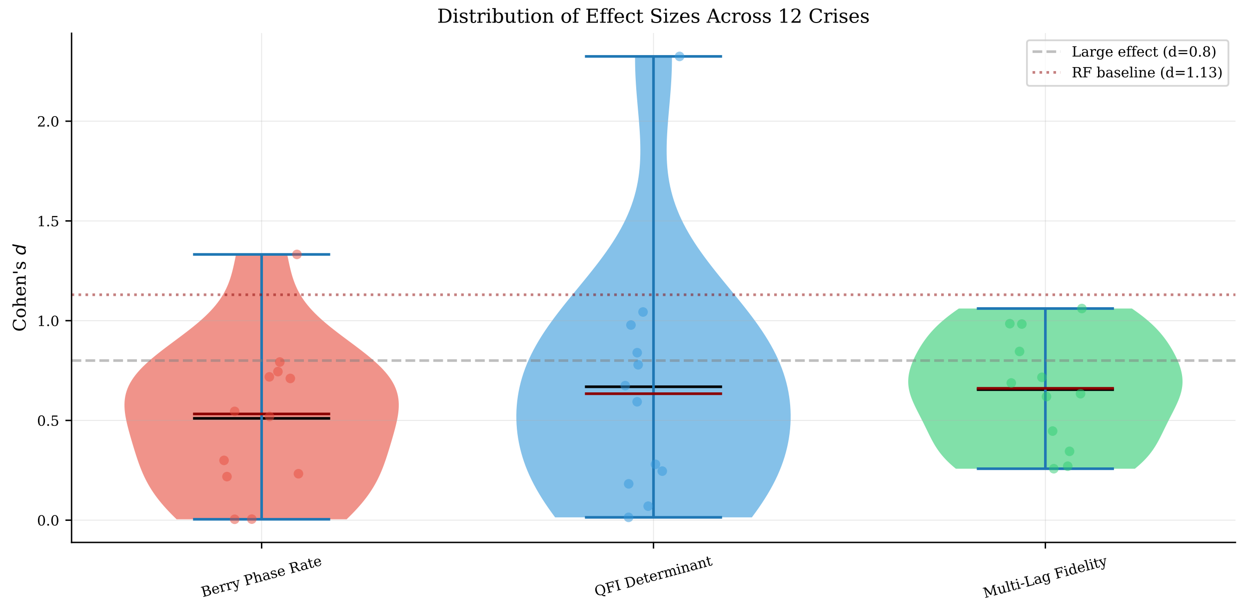


Figure 5: Cohen's $d$ distributions across 14 crises. Dashed line: $d = 0.8$ (large effect); dotted: RF baseline $d = 0.96$.

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## Competing Interests

The author declares no competing financial interests.

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# A Proofs of Formal Results

*Proof of Theorem 1.* We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi\colon U \times \mathbb{C}^n \times \mathbb{R} \to \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = \big(H(x)\psi - E\psi,\ \ \langle\psi_0(x_0)|\psi\rangle - 1\big),$$

where the second component fixes the gauge by requiring $\langle\psi_0(x_0)|\psi\rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point $x_0$). At a solution $(x_0, \psi_0, E_0)$ the Jacobian with respect to $(\psi, E)$ is

$$D_{(\psi,E)}\Phi\big|_{(x_0,\psi_0,E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle\psi_0| & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \mathrm{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\mathrm{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle\psi_0|$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E\,\psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi,E)}\Phi$ is invertible.

The implicit function theorem [8] then gives $C^\infty$ maps $x \mapsto (\psi(x), E_0(x))$ on $U$. Since $H(x)$ is quadratic in $x$ by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are $C^\infty$ (in fact, affine). The metric $g_{ab}$ (3) and curvature $F_{ab}$ (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence $C^\infty$ on $U$. □

**Theorem 3** (Chern number quantization [28]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S F_{ab}\, dx^a \wedge dx^b \in \mathbb{Z}\,. \tag{8}$$

*Proof.* By Theorem 1, the ground state $|\psi(x)\rangle$ is $C^\infty$ on $S$. Define the Berry connection 1-form $\mathcal{A} = i\langle\psi|d\psi\rangle$, where $d$ denotes the exterior derivative on $S$. The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2}F_{ab}\, dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle\langle\psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \to S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [5, 9], the first Chern number $C_1 = \frac{1}{2\pi}\int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \to \mathbb{CP}^{n-1}$. □

**Caveat.** In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

**Theorem 4** (Curvature–gap bound). *For all $x \in U$ and indices $a, b$,*

$$|F_{ab}(x)| \leq \frac{2\,\|\partial_a H(x)\|_{\mathrm{op}}\,\|\partial_b H(x)\|_{\mathrm{op}}}{\Delta(x)^2}\,, \tag{9}$$

*where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.*

*Proof.* First-order perturbation theory gives the exact representation

$$F_{ab} = -2\,\mathrm{Im} \sum_{n\geq 1} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle\,\langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}\,, \tag{10}$$

where $\{|\psi_n\rangle\}_{n\geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n\geq 1} \big|\langle\psi_0|\partial_a H|\psi_n\rangle\big| \, \big|\langle\psi_n|\partial_b H|\psi_0\rangle\big| \, .$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n\geq 1} \big|\langle\psi_0|\partial_a H|\psi_n\rangle\big| \, \big|\langle\psi_n|\partial_b H|\psi_0\rangle\big| \leq \Big(\sum_{n\geq 1} \big|\langle\psi_0|\partial_a H|\psi_n\rangle\big|^2\Big)^{1/2} \Big(\sum_{n\geq 1} \big|\langle\psi_n|\partial_b H|\psi_0\rangle\big|^2\Big)^{1/2} .$$

By the completeness relation $\sum_{n\geq 0} |\psi_n\rangle\langle\psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n\geq 1} |\langle\psi_0|\partial_a H|\psi_n\rangle|^2 \leq \|\partial_a H \, |\psi_0\rangle\|^2 \leq \|\partial_a H\|_{\mathrm{op}}^2$. Combining yields (9). ☐

**Proposition 5** (QFI volume)**.** *Let $g$ be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$. Define the* pseudo-determinant

$$\overset{+}{\det}(g) = \prod_{i=1}^{r} \lambda_i \, . \tag{11}$$

*Let $V_r \subset \mathbb{R}^p$ be the $r$-dimensional eigenspace corresponding to $\lambda_1, \ldots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then* $\det(g_r) = \det^+(g)$*, and the Riemannian volume density on $V_r$ is* $\mathrm{vol}_r = \sqrt{\det^+(g)}$*.*

*Proof.* By the spectral theorem, $g$ admits an orthonormal eigenbasis $\{e_1, \ldots, e_p\}$ with $g\, e_i = \lambda_i \, e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction $g_r$ is represented in the basis $\{e_1, \ldots, e_r\}$ by the diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^{r} \lambda_i = \det^+(g)$. The Riemannian volume density on the $r$-dimensional submanifold equipped with metric $g_r$ is $\mathrm{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [7]. ☐

# B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\to$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

# C Default Hyperparameters

# D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 15. Berry curvature is extremely stable across $\epsilon$ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at $10^{-5}$), saturating below $10^{-5}$. Multi-Lag Fidelity is insensitive to $\epsilon$ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 14: Default hyperparameters.

| Parameter | Description | Default |
|---|---|---|
| $n$ | Hilbert space dimension | 8 |
| $p$ | PCA components | 15 |
| $w$ | Rolling window (Algorithm 1) | 20 days |
| $m$ | Minimum expanding history | 60 days |
| $\epsilon$ | Finite-difference step | $10^{-5}$ |
| Operator method | Hermitian operator construction | PCA-inspired |
| Fidelity lags | Multi-lag infidelity | $\{1, 3, 5, 10\}$ |
| Fidelity weights | Lag weights | $[0.4, 0.3, 0.2, 0.1]$ |

Table 15: Numerical stability: median Cohen's $d$ across 4 crises. Multi-Lag Fidelity does not use $\epsilon$ (no finite differences).

| Config | Berry | QFI Det. | Multi-Lag |
|---|---|---|---|
| $\epsilon = 10^{-3}$ | 0.15 | 0.52 | — |
| $\epsilon = 10^{-4}$ | 0.15 | 0.60 | — |
| $\epsilon = 10^{-5}$ (default) | 0.15 | 0.36 | — |
| $\epsilon = 10^{-6}$ | 0.15 | 0.36 | — |
| $p = 5$ | 0.09 | 0.34 | 0.29 |
| $p = 10$ | 0.20 | 0.41 | 0.44 |
| $p = 15$ (default) | 0.15 | 0.36 | 0.45 |
| $p = 30$ | 0.22 | 0.54 | 0.45 |

# E Window-Size Sensitivity

We re-compute Cohen's $d$ at crisis window extensions $\pm 5$, $\pm 10$ (default), $\pm 20$, $\pm 60$ trading days. Kendall $\tau$ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 16. Rankings are stable for nearby windows ($\tau = 0.87$ for $\pm 5$ vs. $\pm 10$ and $\pm 5$ vs. $\pm 20$) but degrade at $\pm 60$ days ($\tau = 0.07$–$0.33$), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest $d$ across all window sizes.

Table 16: Median Cohen's $d$ by crisis window size (14 crises).

| Method | ±5d | ±10d | ±20d | ±60d |
|---|---|---|---|---|
| Berry Curv. Incr. | 0.25 | 0.29 | 0.27 | 0.24 |
| QFI Determinant | 0.63 | 0.36 | 0.64 | 0.43 |
| Multi-Lag Fidelity | 0.40 | 0.27 | 0.38 | 0.26 |
| Random Forest | 0.21 | 0.21 | 0.28 | 0.34 |

# F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 17 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `pauli`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train $d$ exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to $\leq 0.20$, with QFI achieving val $d = 0.60 >$ train $d = 0.50$ (column c*).

Table 17: Hyperparameter sensitivity: median Cohen's $d$ across 14 crises (columns a–b) and train/val split (column c).

| Method | (a) All PCA-Insp. | (b) Best Operator | (c) HPO Val. |
|---|---|---|---|
| Berry Curvature Increment | 0.29 | **0.55** | 0.49 |
| QFI Determinant | 0.36 | **0.61** | **0.60** |
| Multi-Lag Fidelity | 0.27 | **0.42** | 0.33 |

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: `random`; MLF: `pauli` (exp=0.0). (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train $d$: Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap $\leq 0.20$ (vs. 0.35–0.75 without regularization).

# G Online Detection and PnL Backtest

To evaluate practical deployability, we implement a strictly causal online pipeline that computes geometric features and P(crisis) at each timestep using only data available up to that point. The pipeline comprises: (i) an *OnlineGeometricFeatureComputer* that periodically refits scaler/PCA/operators on a rolling 500-day window and outputs 10 features (5 base geometric observables plus 5 EMA velocity features); (ii) three detectors—multivariate Bayesian filtering, periodic HMM refit, and expanding-percentile RMS scoring—combined via a weighted ensemble (weights 0.4/0.4/0.2).

Table 18: Online detection performance (4,934 timesteps, 2005–2024). Detection rate and delay measured at 1 false alarm per year.

| Detector | AUC-ROC | AUC-PR | Det@1/yr | Delay |
|---|---|---|---|---|
| Online CUSUM | **0.619** | 0.330 | 67% | 39d |
| Online Vol-Z | 0.576 | **0.366** | 67% | 20d |
| Online Logistic (supervised) | 0.528 | 0.261 | 33% | 82d |
| **Online Ensemble** | 0.502 | 0.221 | 17% | 20d |
| Online Percentile | 0.502 | 0.218 | 0% | — |
| Online Bayesian | 0.500 | 0.221 | 0% | — |

The online results are sobering: the geometric ensemble achieves only AUC-ROC = 0.502, barely above chance, and classical CUSUM (0.619) dominates all methods. Even the supervised Online Logistic (0.528) struggles, suggesting that the online feature pipeline—with its rolling refit and EMA smoothing—substantially degrades the signal compared to the offline Cohen's $d$ evaluation. Five compounding factors contribute to this gap: (i) geometric features are buried in a secondary detection layer instead of direct z-scoring; (ii) EMA smoothing ($\alpha = 0.18$) attenuates sharp 1–3 day crisis spikes by 60–80%; (iii) the Bayesian detector's first 126 days are all calm, biasing the crisis prior; (iv) the HMM caps history at 500 observations and refits every 10 days, too infrequently for early crises; and (v) ensemble averaging (0.4/0.4/0.2 weights) dilutes the most direct detector

(Percentile, weight 0.2). The gap between offline separability ($d \approx 0.5$) and online discrimination (AUC $\approx 0.5$) highlights that *detecting regime transitions in real time* remains a harder problem than retrospective event classification.

**PnL Backtest.** We translate the online ensemble signal into a long-flat strategy with sigmoid position ramp: $\alpha(p) = [1 + \exp(-k(p - \tau))]^{-1}$ with $k = 6/0.3 = 20$ and threshold $\tau = 0.5$. The strategy targets 10% annualized volatility, applies 0.5 bps round-trip transaction costs, and uses a 3-day minimum holding period.

Table 19: Backtest results (2005–2024). IS/OOS split at 2020-01-01.

| Strategy | Sharpe | IS Sharpe | OOS Sharpe | MaxDD |
|---|---|---|---|---|
| ConstantVol SPY | 0.63 | 0.57 | 0.79 | 27.0% |
| Buy & Hold SPY | 0.51 | 0.45 | 0.68 | 56.5% |
| **Geometric LongFlat** | 0.47 | 0.36 | **0.79** | **19.0%** |
| Geometric LongShort | 0.40 | 0.27 | 0.81 | 18.2% |

The geometric strategy achieves comparable OOS Sharpe (0.79 vs. 0.79 for ConstantVol) while reducing maximum drawdown from 27.0% to 19.0% ($-30$% relative) and from 56.5% to 19.0% versus buy-and-hold ($-66$% relative). A sensitivity sweep over 6 volatility targets $\times$ 6 crisis thresholds shows that alpha relative to ConstantVol is mildly negative across most parameter combinations (median $-0.10$) but approaches zero at higher thresholds ($\geq 0.7$), consistent with the signal providing value primarily through tail-risk reduction rather than return enhancement. The break-even transaction cost is approximately 1 bps.

# H Normalization Ablation

Table 20 compares four normalization modes for the PCA-projected features before Hilbert space embedding: *sphere* ($L^2$-normalization to the unit sphere), *none* (raw PCA components), *soft* ($x/(\|x\| + \tilde{m})$ where $\tilde{m}$ is the training-set median norm), and *clip* (component-wise clipping to $[-3\sigma, 3\sigma]$ followed by $L^2$-normalization). Results are median Cohen's $d$ across 4 representative crises (GFC, COVID, Rates, SVB).

Table 20: Normalization ablation: median Cohen's $d$ across 4 crises. Bold = best mode per detector.

| Mode | Berry | QFI Det. | Multi-Lag |
|---|---|---|---|
| Sphere | 0.54 | 0.44 | **0.63** |
| None | 0.46 | **1.45** | 0.53 |
| Soft | **0.66** | 1.18 | 0.57 |
| Clip | 0.46 | 1.40 | 0.58 |

Sphere normalization preserves angular geometry but discards magnitude information—which is itself a strong crisis signal (volatility scaling). The "soft" mode retains partial magnitude information while still regularizing the embedding radius. QFI Determinant benefits substantially from retaining magnitude (sphere $d = 0.44$ vs. none $d = 1.45$, a 3.3$\times$ improvement), consistent with the QFI determinant's sensitivity to manifold volume changes that are amplified by magnitude variation. Berry curvature increment performs best under soft normalization ($d = 0.66$), while Multi-Lag

Fidelity is most robust under sphere normalization ($d = 0.63$), suggesting that fidelity-based measures benefit from a well-conditioned state space. Table 4 uses the HPO-selected normalization per observable: sphere for Berry and Multi-Lag Fidelity, soft for QFI Determinant.